

Learn to Denoise

DATA 443/643 · Week 12, note 12.2

Week 12

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Ideas from Stanford CS231N 2025 L14 (predict the noise; T reverse steps). Original slides; that course is not copied.

Lecture plan

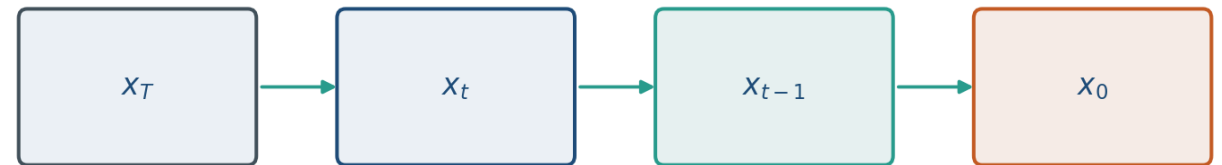
- 1 MSE on ϵ , not an adversary
- 2 One reverse-step mean
- 3 Oracle: true ϵ peels back to x_0
- 4 T net calls versus one GAN pass

Key goal: the net predicts the noise you added in 12.1.

Ordinary regression

- Sample t , form x_t , guess ε .
- Predicting ε is equivalent (up to scale) to x_0 or the score.
- t is an input. The same x_t is a different problem at $t=10$ and $t=900$.

Reverse: a network predicts the noise (or x_0) and steps toward data



DDPM training loss

$$\mathcal{L} = \|\varepsilon - \varepsilon_{\theta}(x_t, t)\|^2$$

No discriminator. A known target.

Sampling still walks $t = T \dots 1$.

Lab 12 uses the true posterior mean.
You know x_0 .

Reuse $x_t = 2.0$ from 12.1

Oracle $\varepsilon_\theta = \varepsilon = 2$: one reverse mean lands on $x_0 = 1$.

- $\alpha_t = \bar{\alpha}_t = 0.64$, $\varepsilon_\theta = 2$.
- The reverse mean lands on $x_0 = 1.0$.
- If $\varepsilon_\theta = 0$, you mostly rescale x_t . Not a sample.

piece	value
x_t	2.0
$(1 - \alpha_t)/\sqrt{1 - \bar{\alpha}_t} \cdot \varepsilon$	1.2
x_{t-1} mean	1.0

The trade

GAN (Week 11)

One forward pass. Min-max. Collapse is the usual failure.

DDPM

T network calls (or fewer with DDIM). Report the sampler next to the metric.

What to take from this lecture

Oracle

Lab 12 does not train a U-Net. It uses the known x_0 .

Noise on the way down

Drop $\sigma_t z$ and every sample looks the same.

Report T

Changing T at test time is part of the method.